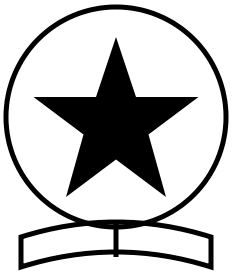
TEXAS STAAR



GRADE 8

Reading Language Arts & Mathematics

All-in-One Study & Practice Guide

8 Full-Length Practice Tests · Complete Skill Review · Deep Answer Explanations

Short & Extended Constructed Response Mastery · Score Diagnostics

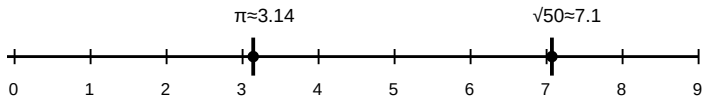
100% aligned to the Texas Essential Knowledge and Skills (TEKS)

2027 EDITION

Andrian Ross

Strand 1 · Numerical Representations and Relationships

Placing irrational numbers on a number line (8.2B)



What this strand tests: how to represent, compare, and order real numbers — including irrational numbers and scientific notation.

Standards covered:

- ○ **8.2A** — Describe relationships between sets of real numbers using a visual representation.
- ○ **8.2B** — Approximate an irrational number (including π and square roots of numbers less than 225) and locate it on a number line.
- ○ **8.2C** — Convert between standard decimal notation and scientific notation.
- ★ **8.2D** — Order a set of real numbers from mathematical and real-world contexts. *(Readiness)*

8.2A · The Real Number System ○

Every number you use at Grade 8 is a **real number.** Real numbers divide into groups nested inside one another:

- **Natural:** 1, 2, 3, … **Whole:** 0, 1, 2, 3, … **Integers:** … −2, −1, 0, 1, 2, …
- **Rational:** any number written as a fraction a/b — integers, terminating decimals (0.75), repeating decimals (0.333…).
- **Irrational:** cannot be written as a simple fraction; decimals never end or repeat — such as **π** and **√2**.

Key Idea: The groups are *nested* — every natural number is also whole, an integer, and rational. But rational and irrational never overlap.

Worked example. Classify √16 and √20. √16 = 4 → natural, whole, integer, **rational.** √20 ≈ 4.472… → **irrational.**

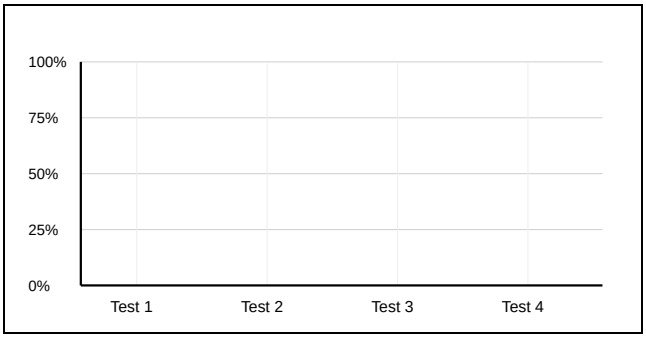
Watch Out: A square root is irrational only when the number is **not** a perfect square. √25 = 5 is rational; √26 is irrational.

Part 4 · Diagnose & Grow

Scoring a test is only step one. The real gains come from seeing **which skills need work** and studying those next. Use these pages after every practice test.

Score-Tracking Sheet

Plot your scores and watch them climb



After each test, put a dot at your percent, then connect the dots.

After each test, count the questions you answered correctly and write it here. Watch your scores climb from Test 1 to Test 4.

Test	Date	Points Earned	Out Of	Estimated Level
Math 1			~48	
Math 2			~48	
Math 3			~48	
Math 4			~48	
RLA 1			~56	
RLA 2			~56	
RLA 3			~56	
RLA 4			~56	

If you earned about…	Estimated level
Less than ~35%	Did Not Meet Grade Level
~35%–55%	Approaches Grade Level